

Practical - 3

Practical Title: Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind

Problem Definition: Design responsive layouts for Home, About, Registration, Dashboard, and Events pages using CSS Grid and Flexbox. Optional to use CSS framework such as Bootstrap or Tailwind. Ensure mobile-first responsiveness.

A. Theory

1. How does the layout adapt for mobile, tablet, and desktop screens?

Answer: The layout adapts to different screen sizes using responsive web design. CSS media queries can be used to apply different styles at different screen widths. On smaller screens such as mobile phones, the content can be arranged vertically, while on tablets and desktops it can use wider layouts with multiple columns. This makes the website usable on mobile, tablet, and desktop devices.

Reference Link:

https://www.w3schools.com/Css/css_rwd_mediaqueries.asp

2. Which layout technique is used and why?

Answer: The CSS Flexbox layout technique is used because it provides a flexible and responsive way to arrange elements in rows or columns. It makes it easier to align items, distribute space, and adapt the layout to different screen sizes. Flexbox is suitable for Student Hub components such as navigation menus, cards, forms, and content sections.

Reference Link:

https://www.w3schools.com/css/css3_flexbox.asp

3. Are typography, spacing, colors, and contrast readable?

Answer: Yes. Readable fonts, suitable font sizes, proper spacing, and sufficient contrast between text and background are used to improve readability. Good color contrast is especially important for users with visual difficulties. Proper font size and line height also make the content easier to read.

Reference Link:

https://www.w3schools.com/css/css_accessibility.asp

4. Is the framework used consistently without unnecessary inline CSS?

Answer: Yes. The website uses a consistent CSS approach by keeping common styling in an external CSS file rather than repeatedly using inline CSS. An external stylesheet allows the same styles to be applied to multiple HTML pages, making the website easier to maintain and keeping the code organized. Inline CSS is avoided unless a specific individual style is required

Reference Link:

https://www.w3schools.com/css/css_external.asp

B. Implementation/ Output/Result

```

C. /* Reset */
D. *{
E.     margin:0;
F.     padding:0;
G.     box-sizing:border-box;
H. }
I.
J. /* Body */
K. body{
L.     font-family:Arial, Helvetica, sans-serif;
M.     background-color:#f4f7fc;
N.     color:#333;
O. }
P.
Q. /* Main Heading */
R. h1{
S.     background-color:#0b5ed7;
T.     color:white;
U.     padding:20px;
V.     text-align:center;
W. }
X.
Y. /* Search Box */
Z. input[type="text"]{
AA.     width:250px;
BB.     padding:8px;
CC.     margin:20px;
DD.     border: 2px solid blue;

```

```

EE.    border-radius:5px;
FF.}
GG.
HH. input[type="password"]{
II.    width:250px;
JJ.    padding:8px;
KK.    margin:20px;
LL.    border: 2px solid blue;
MM.    border-radius:5px;
NN.}
OO.
PP. input[type="email"]{
QQ.    width:250px;
RR.    padding:8px;
SS.    margin:20px;
TT.    border: 2px solid blue;
UU.    border-radius:5px;
VV.}
WW.
XX. input[type="tel"]{
YY.    width:250px;
ZZ.    padding:8px;
AAA.    margin:20px;
BBB.    border: 2px solid blue;
CCC.    border-radius:5px;
DDD. }
EEE.

FFF. input[type="submit"]{
GGG.    width:250px;
HHH.    padding:8px;
III.    margin:20px;
JJJ.    border-radius:5px;
KKK.    background-color: #0b4b90;
LLL.    color: white;
MMM. }
NNN. /* Login and Sign Up Links */
OOO. a{
PPP.    text-decoration:none;
QQQ.    color:#0b5ed7;
RRR.    font-size:18px;
SSS.    margin-right:15px;
TTT. }
UUU.

```

```

VVV.  a:hover{
WWW.    color:red;
XXX.  }
YYY.
ZZZ.  /* Horizontal Line */
AAAA. hr{
BBBB.    margin:20px 0;
CCCC.  }
DDDD.
EEEE. /* Menu Heading */
FFFF. h3{
GGGG.    font-size:35px;
HHHH.    margin-left:20px;
IIII.    margin-bottom:10px;
JJJJ.    color:#0b5ed7;
KKKK.  }
LLLL.
MMMM. /* List */
NNNN. ul{
OOOO.    margin-left:40px;
PPPP.    margin-bottom:25px;
QQQQ.  }
RRRR.
SSSS. li{
TTTT.    margin-bottom:10px;
UUUU.  }
VVVV.
WWWW. /* Welcome Heading */
XXXX. h2{
YYYY.    margin-left:20px;
ZZZZ.    color:#0b5ed7;
AAAAA. }
BBBBB.
CCCCC. /* Paragraph */
DDDDD. p{
EEEE.    margin:20px;
FFFFF.    font-size:18px;
GGGGG.    line-height:1.6;
HHHHH. }
IIIII.
JJJJJ. table{
KKKKK.    width: 700px;
LLLLL.    margin: 40px auto;
MMMMM.    border-collapse: collapse;

```

```

NNNNN.    background-color: white;
OOOOO.    border-radius: 10px;
PPPPP.    overflow: hidden;
QQQQQ.    box-shadow: 0 4px 12px rgba(0,0,0,0.2);
RRRRR.
SSSSS. }
TTTTT.
UUUUU. th{
VVVVV.    color: black;
WWWWW.    padding: 15px;
XXXXX.    font-size: 20px;
YYYYY.    text-align: center;
ZZZZZ. }
AAAAA.
BBBBBB.   td{
CCCCC.    padding: 15px;
DDDDD.    font-size: 18px;
EEEEEE.    text-align: center;
FFFFF.    border: 1px solid #ddd;
GGGGG.    }
HHHHHH.
IIIIII.   td{
JJJJJJ.    padding: 15px;
KKKKKK.    font-size: 18px;
LLLLLL.    text-align: center;
MMMMM.    border: 1px solid #ddd;
NNNNNN.    }
OOOOOO.
PPPPPP.   tr:hover{
QQQQQQ.    background-color: #e3f2fd;
RRRRRR.    }
SSSSSS.
TTTTTT.   th:first-child{
UUUUUU.    border-top-left-radius: 10px;
VVVVVV.    }
WWWWW.
XXXXXX.   th:last-child{
YYYYYY.    border-top-right-radius: 10px;
ZZZZZZ.    }
AAAAAAA.
BBBBBBB.   label{
CCCCCCC.    display: block;
DDDDDDD.    margin-top: 15px;
EEEEEEE.    font-size: 18px;

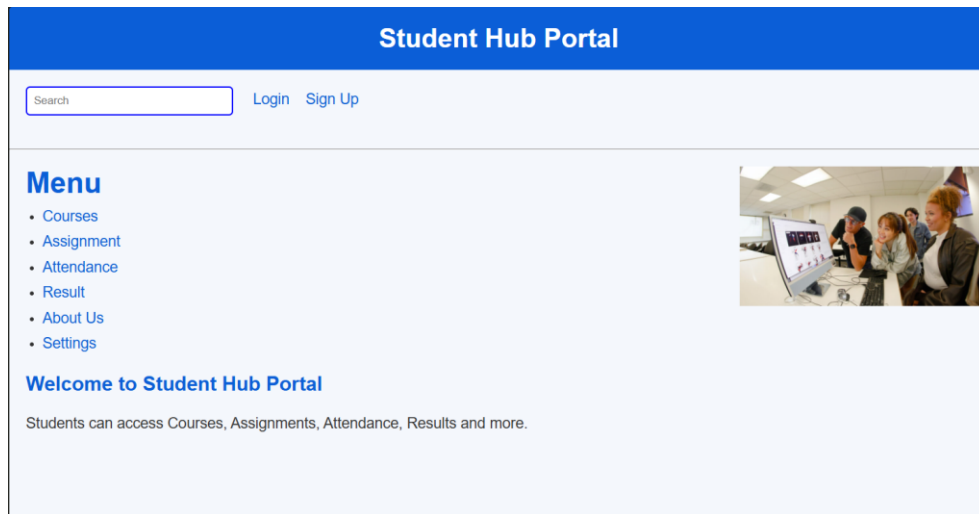
```

```

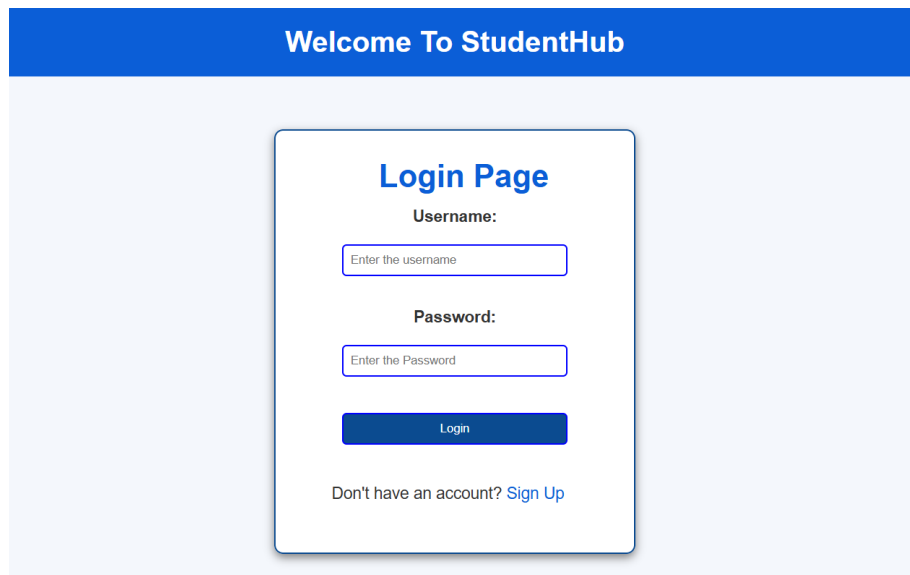
FFFFFFF.      font-weight:bold;
GGGGGGG.
HHHHHHH.    }
IIIIIII.
JJJJJJJ.    input{
KKKKKKK.      width:100%;
LLLLLLL.      padding:10px;
MMMMMMM.      margin-top:5px;
NNNNNNN.      margin-bottom:15px;
OOOOOOO.      border:2px solid blue;
PPPPPPP.      border-radius:5px;
QQQQQQQ.      box-sizing:border-box;
RRRRRRR.    }
SSSSSSS.
TTTTTTT.    fieldset{
UUUUUUU.      width:400px;
VVVVVVV.      margin:40px auto;
WWWWWWW.      padding:30px;
XXXXXXX.      border:2px solid #0b4b90;
YYYYYYY.      border-radius:10px;
ZZZZZZZ.      background-color:white;
AAAAAAA.      box-shadow:0px 4px 10px gray;
BBBBBBB.      text-align:center;
CCCCCCC.    }
DDDDDDD.

```

C. Evidence of Work (Screenshots)

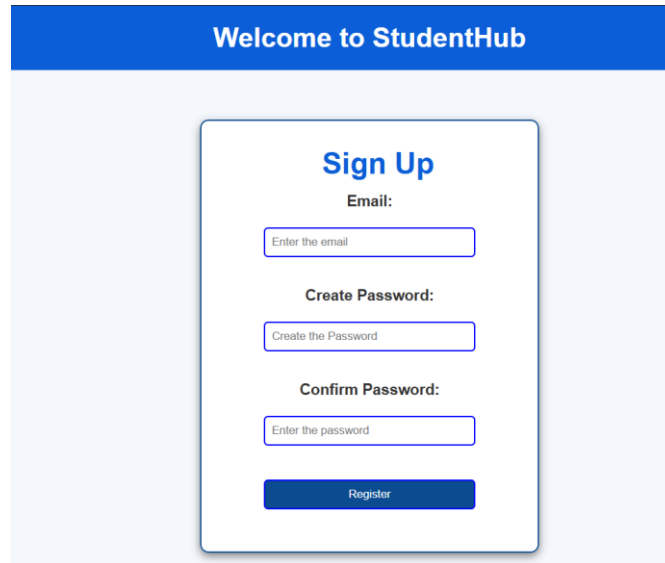


Description: This figure shows the Home Page of the Student Hub Portal. It contains a header with the Student Hub Portal title, a search box, and Login and Sign Up links for user access. The page includes a Menu section with navigation links to Courses, Assignment, Attendance, Result, About Us, and Settings. An image is displayed on the right side, while the welcome section provides a brief introduction to the portal and explains that students can access courses, assignments, attendance, results, and other resources



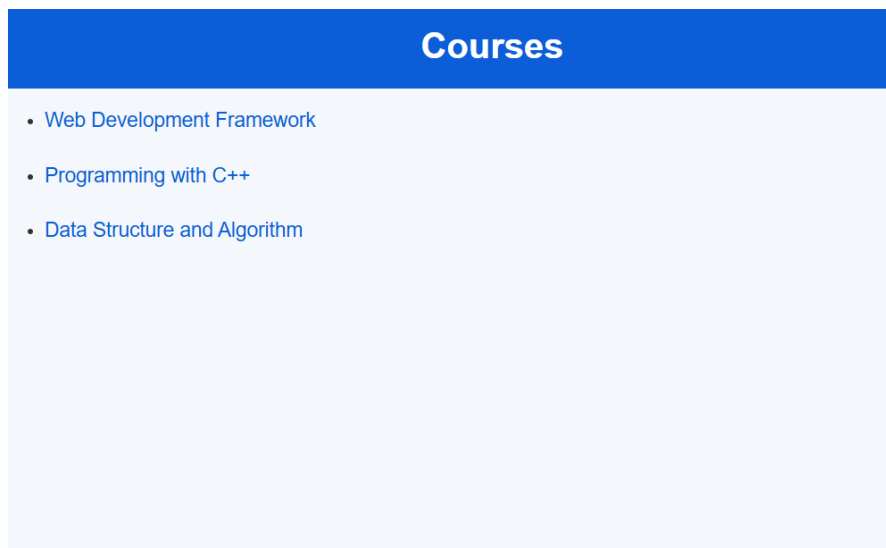
The image shows a web page titled "Welcome To StudentHub" in a blue header. Below the header is a light blue background. In the center is a white login box with a blue border. Inside the box, the title "Login Page" is in blue. Below it, the label "Username:" is in black, followed by a text input field with the placeholder "Enter the username". Below that, the label "Password:" is in black, followed by a text input field with the placeholder "Enter the Password". Below the password field is a blue "Login" button. At the bottom of the box, it says "Don't have an account? [Sign Up](#)".

Description: This figure shows the Login Page of the StudentHub Portal. It contains a welcome heading at the top, followed by a login form with fields for Username and Password. A Login button is provided to submit the credentials. For new users, a Sign Up link is available to create an account. The page uses a simple and clean layout with blue headings, bordered input fields, and a centered login box for easy navigation and user interaction



The image shows a web form titled "Welcome to StudentHub" with a blue header. Below the header is a white box with a blue border containing the "Sign Up" form. The form has three input fields: "Email:" with a placeholder "Enter the email", "Create Password:" with a placeholder "Create the Password", and "Confirm Password:" with a placeholder "Enter the password". A blue "Register" button is at the bottom of the form.

Description: This figure shows the registration page of StudentHub. Students can enter their email, create a password, and confirm the password. A Register button is provided to create a new account.



The image shows a section titled "Courses" with a blue header. Below the header is a light blue box containing a list of three courses: "Web Development Framework", "Programming with C++", and "Data Structure and Algorithm".

Description: This figure displays the Courses section of the portal. It lists the available courses, including Web Development Framework, Programming with C++, and Data Structure and Algorithm. Students can select these courses to access their academic information.

Web Development Framework

Course Name: Web Development Framework

Faculty: Jay Sir

Topics: HTML, CSS, JavaScript

Duration: 12 Weeks

Materials: Notes and PPTs

Description: This figure provides detailed information about the Web Development Framework course. It shows the course name, faculty, topics such as HTML, CSS, and JavaScript, course duration of 12 weeks, and the available learning materials such as notes and PPTs.

Assignments		
Course	Due Date	Status
Web Development Framework	27 July	Pending
Programming with C++	22 July	Pending
Data Structure And Algorithm	14 July	Submitted

Description: This figure shows the Assignments section of StudentHub. Assignments are displayed in a table containing the course name, due date, and submission status. The status indicates whether an assignment is Pending or Submitted.

Attendance	
Subject	Attendance
Web Development Framework	90%
Programming with C++	80%
Data Structure And Algorithm	85%

Description: This figure displays the student's attendance information for different subjects. The table shows the subject names along with their attendance percentages, allowing students to monitor their attendance.

Semester Result	
Subject	Grade
Web Development Framework	A+
Programming with C++	A+
Fundamental Of DSA	A+

Description: This figure shows the Semester Result section. It displays the subjects along with the grades obtained by the student, providing a simple way to view academic performance.

The screenshot shows the 'About Us' section of the StudentHub Portal. It has a blue header with the text 'About Us'. Below the header, there is a section titled 'Welcome to the StudentHub Portal!' in blue. This is followed by a paragraph: 'We are providing the services which are essential for the students like Attendance Record, Course Registration etc.' Below this, there are three sections: 'Our Aim' with the text 'Our Aim is to provide a services wwhich are reliabe,simple and easy to use.', 'Services' with a bulleted list: 'Attendance Record', 'Course Registration', and 'Timetable', and 'Contact Us' with the text 'Email: xyz@gmail.com' and 'Phone: 9863628735'.

About Us

Welcome to the StudentHub Portal!

We are providing the services which are essential for the students like Attendance Record, Course Registration etc.

Our Aim

Our Aim is to provide a services wwhich are reliabe,simple and easy to use.

Services

- Attendance Record
- Course Registration
- Timetable

Contact Us

Email: xyz@gmail.com
Phone: 9863628735

Description: This figure presents the About Us section of the StudentHub Portal. It explains the purpose of the portal, its aim, the services provided to students, and contact information. The services include attendance records, course registration, and timetable.

The screenshot shows the 'Settings' page of the StudentHub Portal. It has a blue header with the text 'Settings'. Below the header, there is a section titled 'Profile Information' in blue. This section contains four form fields: 'Full Name:' with the value 'John Doe', 'Student ID:' with the value 'ST12345', 'Email:' with the value 'john@gmail.com', and 'Phone:' with the value '9876543210'. At the bottom of the section, there is a blue link labeled 'Logout'.

Settings

Profile Information

Full Name:
John Doe

Student ID:
ST12345

Email:
john@gmail.com

Phone:
9876543210

[Logout](#)

Description: This figure shows the Settings page containing the student's profile information. It includes fields for Full Name, Student ID, Email, and Phone Number, along with a Logout option. This page allows students to view or manage their personal information.

D. GitHub Repository Link:

<https://github.com/devanshirshah24-pixel/WDF-Practicals/tree/main/Practical3>

E. Challenges Faced:

During the implementation of the responsive StudentHub layouts, I faced the following challenges:

1. Responsive layout: It was challenging to make the pages look proper on different screen sizes. This was solved by using CSS media queries.
2. Using Grid and Flexbox: Initially, it was difficult to decide when to use Grid and when to use Flexbox. Flexbox was mainly used for navigation and alignment, while Grid was used for card-based and multi-column layouts.
3. Mobile-first design: Designing the layout for smaller screens first required careful planning of the content and spacing.
4. Maintaining consistency: The same typography, spacing, navigation, and design patterns had to be maintained across Home, About, Registration, Dashboard, and Events pages.
5. Avoiding unnecessary inline CSS: Styling was moved to an external CSS file so that the same styles could be reused across multiple pages.

F. Learning Outcomes:

- How to design responsive web layouts.
- How to use CSS Flexbox for flexible element alignment.
- How to use CSS Grid for row and column-based layouts.
- How to use media queries for different screen sizes.
- How to follow a mobile-first approach.

- How to maintain readable typography, spacing, colors, and contrast.
- How to use an external CSS stylesheet instead of unnecessary inline CSS.
- How to create consistent layouts across multiple StudentHub pages.
- How to test responsive designs using the browser's responsive/device mode.
- How responsive design improves the usability of a website on different devices.

Faculty Remarks: (Keep this blank for faculty)